

Experiment 1:

Aim: To demonstrate confusion matrix using python

Program

```
import numpy as np
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

actual = np.array(
['Dog','Dog','Dog','Not Dog','Dog','Not Dog','Dog','Dog','Not Dog','Not Dog'])
predicted = np.array(
['Dog','Not Dog','Dog','Not Dog','Dog','Dog','Dog','Dog','Not Dog','Not Dog'])
conf_matrix=confusion_matrix(actual,predicted)

sns.heatmap(conf_matrix,annot=True,fmt='g',xticklabels=['Dog','Not Dog'],yticklabels=['Dog','Not Dog'],cmap='RdPu')

plt.ylabel("prediction",fontsize=16)
plt.xlabel("actual",fontsize=16)
plt.title("confusion matrix",fontsize=20)
plt.show()
```

Output:

